

Integration and System Test Report

ExpenseMate – Personal Expense Manager

Software Construction, Assignment 1 | Phase 4: Integration & System Testing | Version 1.0, 27 September 2026

37/37 integration tests pass	31/31 system tests pass (first run 28/31)	12 defects found, 11 fixed	2.03 average complexity (409 functions)
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1. Integration

Bottom-up: DB, then Logic + DB (Supertest on a separate test database, emptied before each test), then UI + Logic through the Vite proxy.

Area	Cases	Passed	Area	Cases	Passed
Categories API	2	2	CSV import	4	4
Transactions API	3	3	Unknown API paths	1	1
Budgets API	1	1	Trend with many transactions on the same day	1	1
Analytics API	1	1	Sign up	3	3
Update transaction API	4	4	Log in and log out	4	4
Delete category API	2	2	Demo login	4	4
Trend API	2	2	Protected data	3	3
CSV export	1	1	GET /api/health	1	1
			Total	37	37

Defects D-05 to D-08 were found at this stage.

2. System Testing

Black-box tests against the SRS: 24 run by script through the web app's /api address, 7 run by hand in a browser, on a separate database. After the three fixes the whole suite was re-run.

ID	Req.	Test case	Expected result	1st run	Final
ST-01	FR-01	Sign up with valid details	201; session cookie is HttpOnly and SameSite=Lax; no password hash returned	Pass	Pass
ST-02	FR-01	Sign up again with the same email in capitals	409 with a message on the email field	Pass	Pass
ST-03	FR-01	Sign up with a short password and an invalid email	400 naming both fields	Pass	Pass
ST-04	FR-02	New account has starter categories	8 categories: Food, Transport, Bills, Shopping, Entertainment, Health, Salary, Freelance	Pass	Pass
ST-05	FR-03	Log out, then log back in	After logout /auth/me is 401; after login it returns the user	Pass	Pass
ST-06	FR-03	Session lasts about 30 days	Cookie expiry 30 days ahead	Pass	Pass
ST-07	FR-05	Add, duplicate and delete a category	201, then 409 for the duplicate, then 204	Pass	Pass
ST-08	FR-06	Record an expense of Rs 1,500.50 with a note	201; stored as 150050 paisa; appears in the September list	Pass	Pass
ST-09	FR-06	Reject invalid transaction input	400 for a negative amount, a fractional amount and a 201-character note	Pass	Pass
ST-10	FR-07	Edit then delete a transaction	PATCH 200 with new amount, DELETE 204, then gone from the list	Pass	Pass
ST-11	FR-05	Try to delete a category that has transactions	409 and the category remains	Pass	Pass
ST-12	FR-08	Filter September expenses	Only September expenses, newest first	Pass	Pass
ST-13	FR-09, FR-10	Budget moves from On track to Close to limit to Over budget	Rs 1,000 Food budget: 85% WARNING, then 105% EXCEEDED and listed in alerts; updating keeps the same budget	Pass	Pass
ST-14	FR-11	Monthly summary matches the transactions	Income 58000.00, expenses 2550.00, balance 55450.00 (computed independently from the list)	Pass	Pass
ST-15	FR-12	Category breakdown adds up	Per-category totals equal the list; shares sum to about 100%; largest first	Pass	Pass

ID	Req.	Test case	Expected result	1st run	Final
ST-16	FR-13	Six-month trend includes empty months	6 rows April to September; April to July are zero; September equals the summary	Pass	Pass
ST-17	FR-14	Export September as CSV	text/csv attachment; header plus one row per September transaction; amounts in rupees	Pass	Pass
ST-18	FR-15	Import a valid CSV, then an invalid one	Valid: 3 imported, 1 new category. Invalid: 400 listing lines 3 and 4, nothing saved	Pass	Pass
ST-19	FR-04	A second user cannot see or change the first user's data	Empty list and zero summary; 404 for edit, delete, using the other user's category or budgeting it	Pass	Pass
ST-20	NFR-01	Passwords and session tokens are stored only as hashes	Stored hash is salt:hash and never contains the password; stored token hash is SHA-256 of the cookie, not the cookie itself	Pass	Pass
ST-21	NFR-04	Broken JSON and unknown API paths	Clear JSON error messages, no stack traces	Fail	Pass
ST-22	NFR-05	Response times with 10,000 transactions	Every request under 500 ms (95th percentile); full dashboard data under 2 s	Fail	Pass
ST-23	NFR-02	Same message for wrong email and wrong password	Both 401 with identical text	Pass	Pass
ST-24	NFR-02	Repeated failed logins are blocked	429 after at most 20 failed attempts in 15 minutes	Pass	Pass
ST-25	FR-03	Log in through the login form	Dashboard opens with a personal greeting and the same balance the API reports	Pass	Pass
ST-26	NFR-06	Keyboard shortcut N	New transaction dialog opens with the amount field focused; Escape closes it	Pass	Pass
ST-27	FR-06, NFR-06	Record an expense through the form	Saved with a confirmation; dashboard totals update without reloading	Pass	Pass
ST-28	NFR-07	Phone layout at 375 px on all four pages	No sideways scrolling; bottom navigation visible on screen	Fail	Pass
ST-29	NFR-07	Keyboard focus is visible	Every focused control shows a clear outline	Pass	Pass
ST-30	NFR-07	Text contrast against the background	At least 4.5:1 (WCAG AA) for all text colours	Pass	Pass
ST-31	NFR-04	Server stops and restarts while the app is open	Offline banner appears; the app reconnects on its own and the user stays logged in	Pass	Pass

First run 28/31: ST-21, ST-22 and ST-28 failed (D-09, D-10, D-11). After fixes: 31/31. Every functional requirement is covered; NFR-09 was verified on Windows only.

3. Performance (NFR-05, target 500 ms)

Request (10,000 transactions)	p95 before	p95 after
Transactions of one month	290 ms	76 ms
Monthly summary	28 ms	20 ms
Category breakdown	29 ms	16 ms
Six-month trend	523 ms	22 ms
Budgets with status	35 ms	17 ms
CSV export (one month)	451 ms	60 ms
Whole dashboard (worst)	383 ms	102 ms

4. Bug Report

ID	Severity	Found by	Defect	Status
D-01	High	Unit test	Wrong Date constructor; monthly queries crashed	Fixed
D-02	Critical	Code review	Import of a missing file; server would not start	Fixed
D-03	High	Code review	Export name mismatch; listing crashed	Fixed
D-04	Medium	Code review	Error handler mounted too early; HTML 500 pages	Fixed
D-05	High	Integration	Wrong enum value; budgets API returned 500	Fixed
D-06	High	Integration	Transaction DELETE route missing :id	Fixed
D-07	High	Integration	Wrong table in count; category delete crashed	Fixed
D-08	High	Integration	Category DELETE route missing :id	Fixed
D-09	Low	System ST-21	Unknown API paths returned HTML, not JSON	Fixed
D-10	Medium	System ST-22	Trend took 523 ms with 10,000 rows; now totals in the database (22 ms)	Fixed
D-11	High	System ST-28	Phone navigation off-screen after the login change	Fixed
D-12	Medium	System ST-24	Rate limiter would treat all users as one behind a hosting proxy	Open (deployment)

5. Fault-Find and Fault-Fix Rates

Measure	Value
Fault-fix rate	91.7% (11 of 12); the open one affects hosting only
Time to fix	Same day, each with a regression test
Defect density	3.1 per KLOC (12 defects, about 3.9 KLOC)
Found by automated tests	75% (9 of 12)
First-run system pass rate	90.3%



Figure 1. Defects by the activity that found them

6. Complexity Metrics

Cyclomatic complexity (McCabe) per function, measured with the ESLint 9 *complexity* rule. Guide: 1–10 acceptable, above 10 complex.

Module	Functions	Average	Highest	Above 10
UI	292	2.24	25	7
Logic	84	1.64	5	0
DB	33	1.18	4	0
All	409	2.03	25	7

Complexity	Function	File
25	TransactionForm	context/TransactionModalContext.jsx
17	TransactionRow	components/TransactionRow.jsx
15	Transactions	pages/Transactions.jsx
13	request	lib/api.js
13	Hero	pages/Dashboard.jsx
12	Budgets	pages/Budgets.jsx
12	Categories	pages/Categories.jsx

Server code stays simple (max 5). All complex functions are React components, where each conditional in the markup counts as a branch; they are the refactoring targets for Phase 5.